

Three-gap of the fold mask, refuted

Steinhaus and log-local SATZ, the ν -mask is not three-gap, and $n_{\text{uniq}} \leq 3$
does not close F_ε

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Abstract

Rounds 393–394 reduced both remaining analytic rests ($\Delta^2 \log \tau$ under F_1 , and Assist) to *occupation regularity* of the ν -mask on the cosine grid: not run-lengths, not checkerboard signs, not equidistribution. The candidate was the Steinhaus three-gap theorem for $\{n\alpha\}$, locally for folded $\log(p^k)$ via the f'' of $\log$. What is proved: Steinhaus on $\mathbb{Z}/q$ (exact, 309/309 Farey and Fibonacci windows) with the additive law $a + b = c$ when three gaps occur; PNT-free integer-log local three-gap for $n \geq 512$, $M \leq 64$. What is *refuted*: that the ν -mask of a canonical window is globally or blockwise three-gap (FRAME-A $n_{\text{uniq}} = 12$; core-42 in [8, 18], none ≤ 5 ; EXT n_{uniq} grows with S), and that $n_{\text{uniq}} \leq 3$ plus F_1 plus $\rho_{\text{AP}} < 1/5$ implies the $\Delta^2 \log \tau$ box (wrapping random $(1, 2, 3)$ -gap F_1 -masks leave every legal JUMP at $h = 40$; mechanical φ misses the box at $h = 80$). The 3/8-floor and MED-CAP 8/3 are not shadows of the cosine-grid gap spectrum. No RH claim.

Status. Lemmas 1–2 are proved (finite identities / a named range). Propositions 1–2 name the refutation. No RH claim. No L^* claim.

1 Recollection

Occupied Fejér deletes the ν -support Ξ from the discrete Bernstein–Szegő cosine grid [2, 3]. F_1 ($\max \nu\text{-run} \leq 2$) is necessary and not sufficient [2]: periodic tiles 1010/1100/110 stay inside $|\gamma - 1/4| \leq 1/16$, a random F_1 half-filling jumps 1.18. Sign-Schur is a coin [1]. The remaining object is the occupation of Ξ itself.

The fold construction places $\log(p^k)$ on a circle of length $L = 4h - 2$ at angle $2\pi k/L$ [5] and folds $j \mapsto \min(j, L - j)$ onto the cosine grid. Sequences $\{n\alpha\}$ have, by Steinhaus, at most three distinct circular gaps. The folded log sequence is not a pure $\{n\alpha\}$, but is locally log-linear ($\log(n+1) - \log(n) \approx 1/n$).

2 Steinhaus, and log-local three-gap

Lemma 1 (Steinhaus on $\mathbb{Z}/q$). *Let p, q be coprime positive integers. For every $N = 1, \dots, q$ the circular gaps of $\{0, p, 2p, \dots, (N-1)p\} \bmod q$ take at most three values. Machine identity on every coprime pair with $q \leq 30$ and on the Fibonacci convergents with $q < 500$ (309/309). When three values $a < b < c$ occur, $a + b = c$ (witnessed on $\{k\varphi\}$ at $N = 4$).*

A random 20-point subset of a 40-point Steinhaus set on $\mathbb{Z}/89$ has five gap lengths: thinning does *not* preserve three-gap. Prime-power selection from a log quasi-AP is a thinning.

Lemma 2 (integer-log local three-gap). *Let $L = 800$ and $x_n = \lfloor L \{\log_2 n\} \rfloor$. For every $n_0 \geq 512$ and every block length $M \leq 64$, the circular gaps of $\{x_n : n_0 \leq n < n_0 + M\}$ take*

at most three values (12/12). The onset is $n_0 = 128$, $M = 8$ ($n_{\text{uniq}} = 3$). For $n_0 \leq 64$, $M = 32$ one has $n_{\text{uniq}} \in \{8, 17, 18, 19\}$: fast drift, a finite census, not a SATZ. The argument is PNT-free: it uses only that $(\log)'' = -1/n^2$ is small at large n .

Small- n blocks of the log skeleton therefore remain a named census. They are the honest exception list of Lemma 2.

3 The mask is not three-gap

Write n_{uniq} for the number of distinct *linear* gaps of the occupied cosine-grid indices.

Proposition 1 (mask spectrum). *On FRAME-A $w = 9$ ($S = 367$, $n_\nu = 104$): $n_{\text{uniq}} = 12$, gap-1 count = 2 (the two F_1 pairs), $d_{\min}/d_{\text{med}} = 1/3 < 3/8$, $d_{\max}/d_{\min} = 23 > 8/3$. Sliding windows of 12 occupied nodes are three-gap in fraction 0.45 and five-gap in fraction 0.74 (partial, not a block SATZ). Spatial dyadic $L = 4$: nine blocks, all ≤ 5 , seven ≤ 3 ; shifting the cut by $S/32$ moves the counts. The pre-fold circle has the same $n_{\text{uniq}} = 12$. Core-42: $n_{\text{uniq}} \in [8, 18]$ (median 13), 0/42 with $n_{\text{uniq}} \leq 5$, sliding-12 three-gap fraction in $[0.31, 0.60]$. EXT $kz = 97$: $n_{\text{uniq}} = 13$; larger EXT windows reach 20–27 (grows with S). Dead χ_3/χ_4 sample: $n_{\text{uniq}} \in [10, 15]$.*

The 3/8-floor of the mean sieve [6] and MED-CAP 8/3 live on the *log-sieve block packing*, not on the cosine-grid gap spectrum of Ξ . They are not shadows of three-gap.

4 The implication fails

Proposition 2 (discriminator). *On the $h = 40$ cosine grid, wrapping concatenations of $\{1, 2, 3\}$ (an F_1 -legal three-gap mask with $\rho_{\text{AP}} = 0.194 < 1/5$) leave the box (jump 2.33). Random F_1 has $n_{\text{uniq}} = 5$ and jump 7.03. Periodic 1010 ($n_{\text{uniq}} = 1$, $\rho_{\text{AP}} = 1$) and 1100 ($n_{\text{uniq}} = 2$) stay in. At $h = 80$: random F_1 jump 1.179 OUT (r393 reproduced); $\{n\alpha\}$ with F_1 and $\rho_{\text{AP}} < 1/5$ can sit in (e.g. $\sqrt{2} - 1$, jump 0.30); wrapping $\{2, 3, 4\}$ (isolated singles) is IN on 8/8 seeds; wrapping $\{1, 2, 3\}$ (random F_1 pairs) is IN on 1/8. Mechanical φ at fill 0.45 misses the box ($d = 0.1016 > 1/16$) even with three-gap and F_1 .*

Thus $n_{\text{uniq}} \leq 3$ is neither necessary (the source mask has 8–18 global gaps and stays in by the r393 census) nor sufficient (random three-gap with F_1 pairs leaves the box). Isolated-single three-gap $\{2, 3, 4\}$ is a different class from the source (which has a median of four gap-1 pairs on core-42, and a sparse tail out to gap 23).

5 Assist and named kills

On $h = 40$ at $n_0 = \lfloor 2N/5 \rfloor = 16$, $\lambda_{\max}(E_{n_0})$ of the Fejér-deleted Gram is 2.01 on random F_1 , 0.997 on wrapping $\{2, 3, 4\}$, and 1.405 on 1010. Two-period $c = 2/3$ has $\lambda_{\max}(E_{22}) = 1.0288 > 1$ [4, 1]. The Assist spectrum separates sparse isolated-single three-gap from random F_1 and from global AP; it does not identify the source class (which has pairs and a tail).

Named kills:

- two-period / 1010: one gap, $\rho_{\text{AP}} = 1$ (non-periodicity);
- random F_1 : many gaps, jump 1.18;
- scramble seed 3: $n_{\text{uniq}} = 15$, sliding-12 three-gap fraction 0.00 (occupation, matching r393's F_1 break);
- dyadic-cut shift: blockwise n_{uniq} moves with the partition (not a SATZ of a canonical cut).

6 What remains

The occupation object is now strictly smaller than “three-gap of the mask”. The source histogram on FRAME-A is 2–3 *dominated* (counts 44 and 27) with two F_1 pairs and a sparse large-gap tail. That is the shape of a prime-power thinning of the log skeleton of Lemma 2, not of a Steinhaus set and not of a random F_1 word. Both F_ε ($\Delta^2 \log \tau$) and Assist still hang on this histogram, not on $n_{\text{uniq}} \leq B$.

Machine check

Standalone: Steinhaus on $\mathbb{Z}/q$; additive $a + b = c$; thinning; log-local large n ; small- n census; 3/8 and 8/3 shadows. Construction: $w = 9$ spectrum, sliding partial, discriminator, scramble, Assist split. Discovery census: `three_gap_mask_probe.py` (`--smoke` / `full`), core-42, χ , EXT $kz = 97$. Exit line: THREE GAP MASK VERIFIED.

Claim boundary

Finite identities for Steinhaus on cyclic groups and a named range for integer-log local three-gap; a named refutation of three-gap as a property of the ν -mask and as a closing implication for F_ε ; named periodic / random- F_1 / scramble / two-period breaks. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *Sign-Schur, reduced*, standalone note, August 2026, `rh/problem/sign_schur.tex`.
- [2] S. Hamann, *The $\Delta^2 \log \tau$ -field under F_1 , reduced*, standalone note, August 2026, `rh/problem/tau_field.tex`.
- [3] S. Hamann, *The G_ε^μ -lemma, reduced*, standalone note, August 2026, `rh/problem/g_eps_mu.tex`.
- [4] S. Hamann, *The coherence-assist lemma, reduced*, standalone note, August 2026, `rh/problem/coherence_assist.tex`.
- [5] S. Hamann, the cosine-mesh SATZ, round 379.
- [6] S. Hamann, *The mean-sieve floor*, round 361 (3/8-floor and MED-CAP 8/3).